

(1)

Electrochemical OR Wet Corrosion

This type of corrosion takes place under moist conditions ~~through~~ through the formation of minute electrochemical cells (galvanic cells). When a conducting liquid is in contact with metal or two dissimilar metals dipped in a solution, this type of corrosion occurs.

Electrochemical corrosion involves

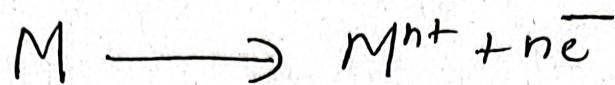
- (1) Formation of anodic or cathodic areas in contact with each other.
- (2) Presence of a conducting medium.
- (3) Corrosion always takes place at anodic area only.
- (4) Formation of corrosion product between anodic and cathodic areas.

Mechanism of WET OR ELECTROCHEMICAL CORROSION

Electrochemical corrosion involves flow of electron current between the anodic and cathodic areas.

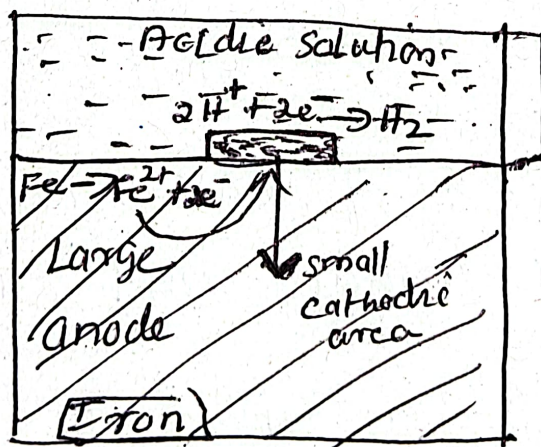
(2)

At Anode: Anode Rx
into ions
Involves dissolution metal

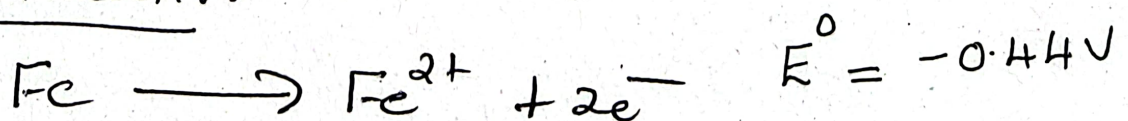


Cathodic Reaction consumes electrons either by (a) evolution of hydrogen or (b) absorption of oxygen, depending on the nature of corrosive environment.

(a) Evolution of hydrogen type corrosion occurs in acidic environment.



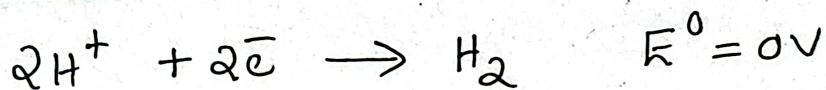
Anode Reaction



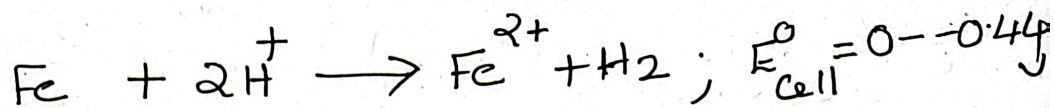
These electrons flow from anode to cathode

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Cathode Reaction



Net Rn



$= 0.44V$

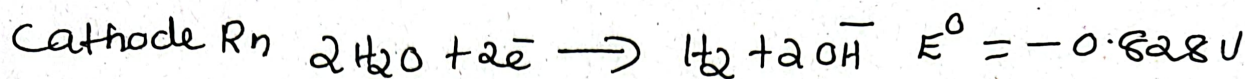
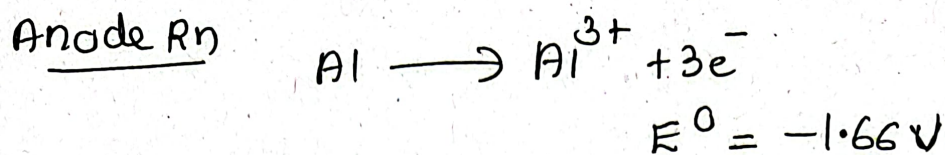
In this type of corrosion displacement of H^+ from acidic solution by the metal takes place. Metal like Fe, Zn, Al etc having higher negative reduction potentials will be anodic and suffer corrosion while metals like Cu, Ag etc having positive reduction potentials will be cathodic and get protection.

Hydrogen Evolution Corrosion in alkaline medium

Hydrogen evolution corrosion does not take place in the case of Iron immersed in alkaline medium. This is due to the fact that hydrogen electrode potential in alkaline medium ($E_{H_2/OH}^0 = -0.828V$) is lower than the electrode potential of Iron ($E_{Fe^{2+}/Fe}^0 = -0.44V$). Whereas metal like aluminium can give hydrogen evolution corrosion in alkaline medium. This is due to the fact that electrode

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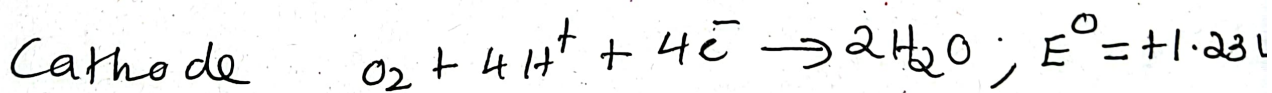
Potential of hydrogen electrode in alkaline media ($-0.828V$) is higher than aluminium $E_{Al^{3+}/Al}^0 = -1.66V$



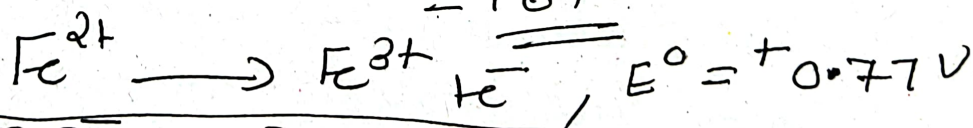
Cell potential $E_{cell}^0 = E_{cathode} - E_{anode}$
 $= -0.828 - (-1.66V)$
 $= 0.832V$

B. Oxygen absorption corrosion in acidic medium

This occurs when metals are in contact with water having dissolved oxygen in acidic or neutral environment eg: Rusting of Iron



$E_{cell}^0 = 1.23 - (-0.44)$
 $= 1.67V$



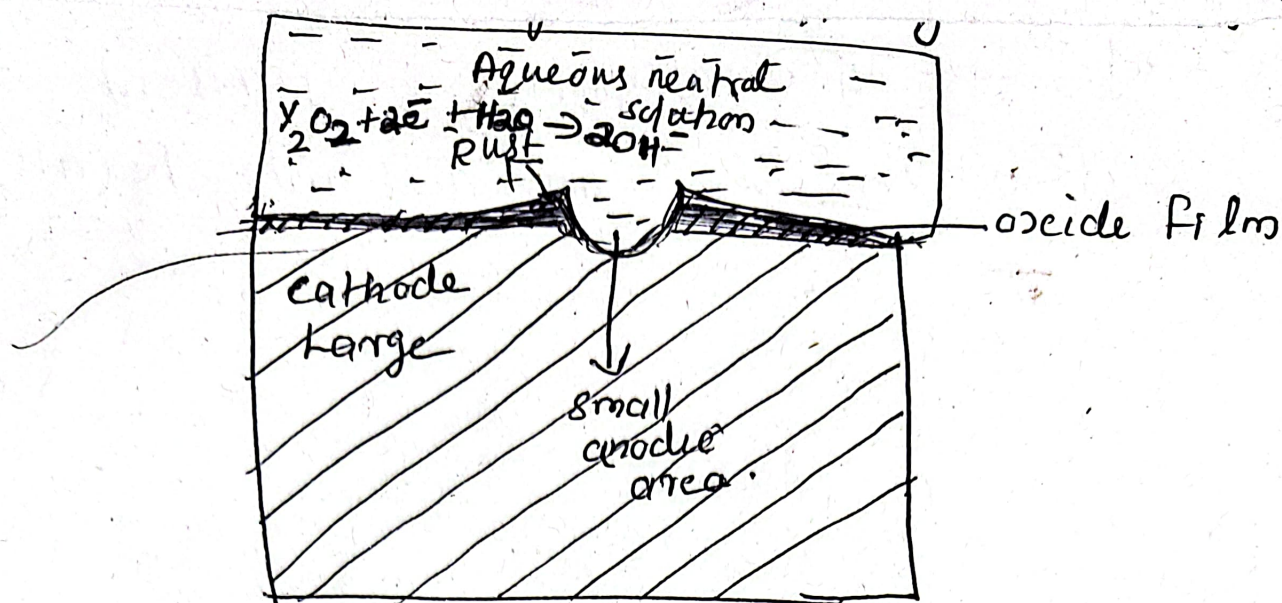
~~$Fe^{2+} + 2OH^- \longrightarrow Fe(OH)_2$~~
~~Fe²⁺ produced forms an insoluble ppt of $Fe(OH)_2$ (yellow rust).~~

Oxygen Absorption corrosion in alkaline

Medium.

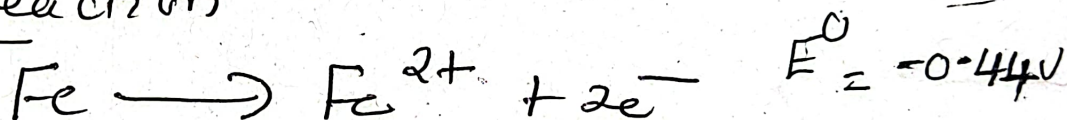
eg: Rusting of Iron in neutral aqueous solution of Electrolyte in the presence of atmospheric oxygen is an example of this type of corrosion.

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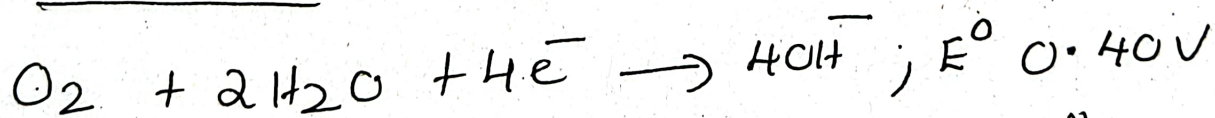
Here crack part act as anode and well metal parts act as cathode.

Anode Reaction



The liberated electrons flow from anode to cathode.

Cathode Reaction



corrosion cell potential in acidic medium $= E_{\text{cath}}^0$

$$E_{\text{cell}}^0 = E_{\text{cathode}} - E_{\text{anode}}$$

$$= 0.40 - (-0.44\text{V})$$

Fe^{2+} is not again oxidised to Fe^{3+} in alkaline environment. Since oxygen electrode

(6)

Potential is less than $E^{\circ}_{Fe^{3+}/Fe^{2+}}$ (0.77V) in alkaline medium.

Galvanic Series The Electrochemical

series gives useful information regarding reactivity of metals, but it does not provide sufficient information in predicting corrosion behaviour. [Galvanic Series is a series in which metals and alloys are arranged in order of their corrosion tendencies.

A metal high in the series is more anodic and undergo corrosion faster.

Corrosion current - per unit area is a parameter to measure corrosion rate of metals and alloys.

Differences between Galvanic and Electrochemical Series

<u>Electrochemical Series</u>	<u>Galvanic Series</u>
1. E° is measured for metals and nonmetals	E° is measured for metals and alloys.
2. SHE - reference Electrode	Calomel electrode - reference electrode
3. E° measured by dipping metals in 1M salt solution	E° is measured by dipping metals and alloys in unpolluted sea water
4. Predicts displacement tendencies	Predicts Corrosion tendencies

Corrosion Control Measurements

Cathodic Protection :-

The Principle involved in this method is to force the metal to be protected to behave like a cathode, thereby corrosion does not occur.

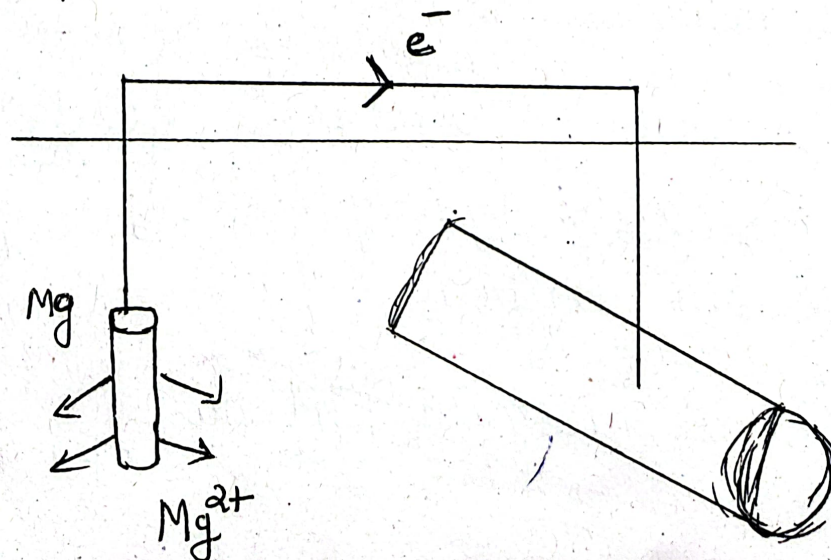
Two Types

1. Sacrificial anodic Protection method

The metallic structure to be protected is connected to a more anodic metal by a wire. So the anodic metal sacrifices and undergoes corrosion. Metal like Mg, Zn and Al are used as sacrificial anode.

Applications

(1) Protection of buried pipe lines, underground cables, marine structures, ship hulls, water-tanks.



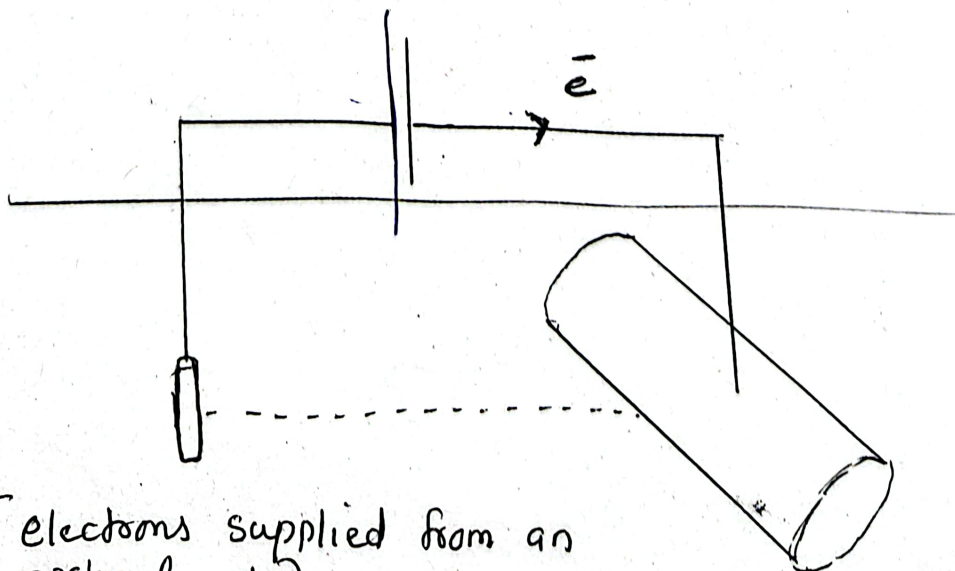
Impressed Current Cathodic Protection

In this method, an impressed current ^{from d.c.} is applied in opposite direction to nullify the corrosion current and convert the metal from anode to cathode. Usually impressed current is derived from d.c. source with an insoluble graphite anode.

Applications

* This type of cathodic protection has been applied to

- (1) Open water--box coolers
- (2) water tanks
- (3) Buried oil and water pipes
- (4) Condensers, transmission line towers



[electrons supplied from an external cells]

Electroless plating ^Q

Electroless plating is a technique of depositing a noble metal on a catalytically active surface of a less noble metal by employing a suitable reducing agent without using electrical energy.



Steps in Electroless plating

1. Preparation of active surface of the object to be Plated.

(a) Etching by acid treatment

(b) Electroplating a thin layer of the metal

(c) Treating with Stannous chloride, followed by dipping in Palladium chloride solution.

(2) Plating Bath consists of

(a) Soluble salt of ^{coating} metal

(b) Reducing agent

(c) complexing agent

(d) Exaltants - enhances the plating rate.

(e) Stabilizers and Buffers.

Advantages of Electroless Plating

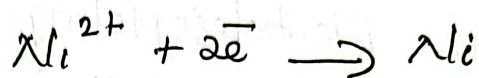
(1) No Electrical Energy is required

(2) Irregular shapes can be plated uniformly

Electroless Ni Plating

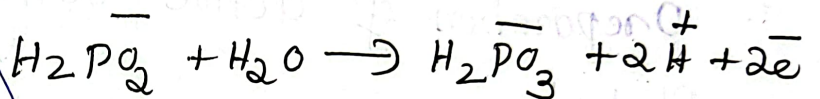
The pretreated object is immersed in the plating bath containing nickel salt and reducing agent like sodium hypophosphite for a certain time and following reactions take place.

At Cathode



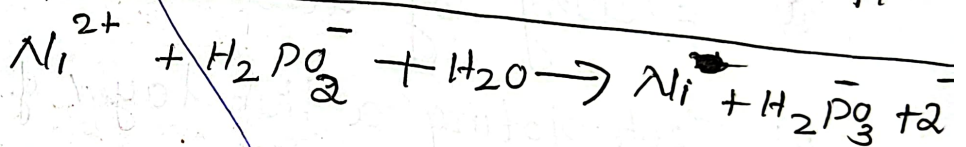
$$E^0 = -0.26\text{V}$$

At Anode



Overall Rn

$$E^0 = 0.499\text{V}$$

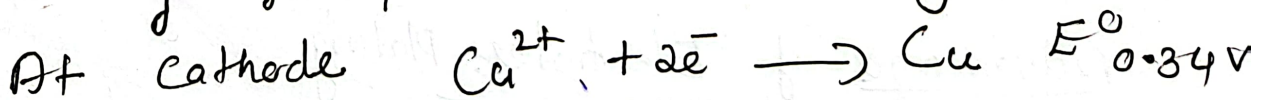


Uses

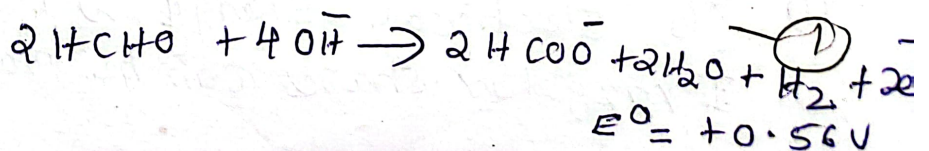
1. In Aerospace components : Engine shafts, engine mounts, compressor blades etc.
2. In Automotive components . For wear protection and corrosion resistance on pistons, gears, ball studs, fuel injectors.

Electroless Copper Plating

The object to be plated is immersed in plating bath containing cupric salt, reducing agents, buffer and complexing agent.



At Anode

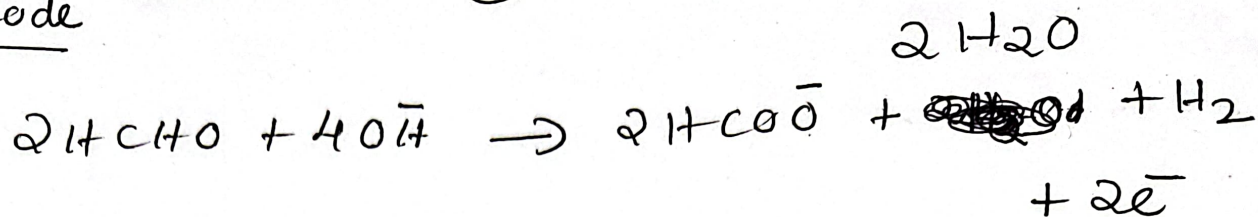


$$E^0 = +0.56\text{V}$$

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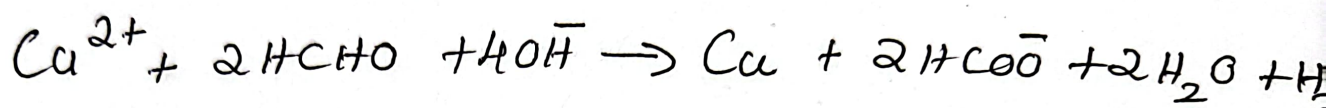
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At Anode



$$E^0 = 0.056V$$

Overall Rn ① + ②



Uses

This type of coating is mainly used in double or multilayered printed circuit boards.

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~~Sign~~